Abitsync Mini-Tasks - PART B

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Why Information Theory?

Information theory is the mathematical study of the quantification, storage, and communication of information. It involves close studying of transfer of bits through a channel, comparing input and output and how they are affected on altering the channel, etc. This concept can be used to study and possibly break code (like the German Enigma engine), recover lost information (using mathematical manipulation based on known properties about the channel), etc.

In this project we will attempt to learn the theoretical basics of Information theory and simulate the working of error control methods.

§5 Expect the Unexpected

You are now *probably* (see what I did there) quite fed up PDFs and PMFs calculations. So let us start a (seemingly) new topic.
All of you (unless you somehow missed out on basic high school education) have heard about averages. The average of some data, in its broadest sense, gives some indication of the overall trend in the data.
Given an experiment and an RV, one might wonder about the average value of the RV over multiple trials of the experiment. It intuitively makes sense that the values of the RV which have a higher probability (PMF or PDF value) will contribute more towards the average than the values which have a very low probability.

With this in mind, we make the following definition,

Definition 5.1. The average or **expectation** of a discrete random variable X is defined as:

$$E[X] = \sum_{\text{all values of } x} x\ p_X(x)$$

This can similarly be extended to continuous random variables as:

Definition 5.2. The average or **expectation** of a continuous random variable X is defined as:

$$E[X] = \int_{\text{all values of } x} x\ f_X(x)\ dx$$

Sadly, as is obvious from our definitions, our old friends PMF and PDF have returned to haunt us. Thankfully, there are certain properties of expectation we can use to avoid them.

Unfortunately for you, we are going to make you derive them.

Problems

1. What can you say about $E[X+Y]$
2. What about $E[aX]$ where a is a constant?

$\sum\sum(x+y)P(x,y) = \sum x P(x,y) + \sum y P(x,y)$

$= \sum x P(x) + \sum y P(y)$

$\sum (ax) P(ax) = a \sum x P(x) = aE(x) = E(x) + E(y)$

$P(ax) = P(x)$ ∵ every element is multiplied by same constant

3. What about $E[g(x)]$? More specifically, what would the reduced formula for that be.

represents mean $\Rightarrow \frac{\int g(x)dx}{\int dx}$

Now that you have a good conceptual understanding of expexctation, here are some more ~~internship~~ problems:

Problems

- Achintya is playing a game where he rolls a die. He can choose to stick to the number obtained or he can choose (only once) to roll again. Help Achintya maximise the value he gets and find the expected value obtained in this case. $1(1/6)+2(1/6)+\cdots 6(1/6) = 21/6 = 3.5$
- An ant is standing on one corner of a cube & can only walk on the edges. The ant is drunk and from any corner, it moves randomly by choosing any edge! What is the expected number of edges the ant travels, to reach the opposite corner?
- A coin is tossed 10 times and the output written as a string. What is the expected number of HH? Note that in HHH, number of HH = 2. (eg: expected number of HH in 2 tosses is 0.25, 3 tosses is 0.5)
- A soda company is holding a contest where everyone who collects one each of N different coupons wins some prize. You get a coupon with each purchase of a soda, and each coupon is equally likely. Find an approximation for the expected number of soda bottles you have to buy in order to collect all the coupons.

§6 It's so Random!!

§6.1 Introduction

Now that you are (hopefully) quite confident regarding random variables, it's time to actually start with the meat of the mini-project.

Definition 6.1. The **entropy** H of a random variable X with a PMF $p_X(x)$ is given as:

$$H(X) = -\sum_{\text{all } x} p_X(x)\log_2 p_X(x)$$

Problems

$E(X) = \sum P_X(x)\cdot x$

$E[-\log_2 P_X(x)] = H(X)$

1. Can you express H in terms of expectation?
2. Can you prove that this entropy is always positive (sad Tenet noises)?

I'd say non-negative

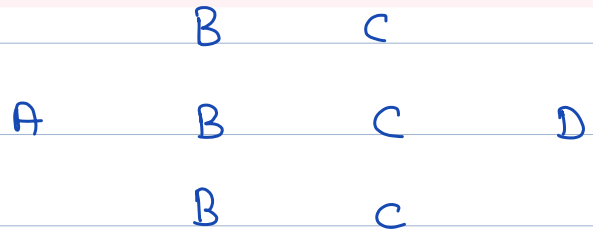
$\log_2 P_X(x) \le 0$ as $P \in [0,1]$. If $P=1 \Rightarrow H=0$

3. Let X be a random variable which takes in values 1, 3, 1888891, 88919819 with probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$ and $\frac{1}{8}$ respectively. Find the entropy of X

$-1/2\log_2(1/2) - 1/4\log_2(1/4) - 1/8\log_2(1/8) - 1/8\log_2(1/8) = 7/4$

As usual, it is only natural to ask whether we can extend entropy to multiple random variables. This, we make the following definition:

An ant is standing on one corner of a cube & can only walk on the edges. The ant is drunk and from any corner, it moves randomly by choosing any edge! What is the expected number of edges the ant travels, to reach the opposite corner?



2 points: $2/3 \times 1/3 = 2/9$ you can do more than 2 points

$S_{BD} = 2/9(2) + 7/9(S_{BD}+2)$ ← rem probability; ↳ 2 extra steps

$\Rightarrow S_{BD} = 9 \Longrightarrow S_{ABD} = 10$

A coin is tossed 10 times and the output written as a string. What is the expected number of HH? Note that in HHH, number of HH = 2. (eg: expected number of HH in 2 tosses is 0.25, 3 tosses is 0.5)

$P(n) = 1/2\, P(n-1) + 1/4\, P(n-1) + 1/4\,(P(n-1)+1)$

n^{th} is tail; n^{th} is head, $(n-1)^{th}$ is tail; n^{th} is head, $(n-1)^{th}$ is head

$P(n) = P(n-1) + \frac{1}{4}$

$= P(n-2) + \left(\frac{1}{4} \times 2\right)$

$= \frac{n-1}{4}$

$P(10) = 2.25$

A soda company is holding a contest where everyone who collects one each of N different coupons wins some prize. You get a coupon with each purchase of a soda, and each coupon is equally likely. Find an approximation for the expected number of soda bottles you have to buy in order to collect all the coupons.

$P = 1/N$

N^{th} Stage: 1 remaining $\Rightarrow P = 1/N \Rightarrow E = N$

$N-1^{th}$ Stage: 2 remaining $\Rightarrow P = 2/N \Rightarrow E = N/2$

i^{th} stage: $N+1-i$ remaining $\Rightarrow P = \frac{N+1-i}{N} \Rightarrow E = \frac{N}{N+1-i}$

1st stage: N remaining $\Rightarrow P = N/N \Rightarrow E = 1$

$\Rightarrow E = N(1 + 1/2 + 1/3 + \cdots + 1/N)$

1. Represent the mutual information expression in terms of summations involving the PDFs only.

$$H(X/Y) = -\sum_{y \in Y} P_Y(y) \sum_{x \in X} P_{X/Y}(x/y) \log \frac{P_{X/Y}(x/y)}{\frac{P_{XY}(x,y)}{P_Y(y)}}$$

$$= -\sum_{y \in Y} P_Y(y) \sum_{x \in X} \frac{P_{XY}(x,y)}{P_Y(y)} \log \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$-\sum\sum P_{XY}(x,y) \log \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$I(X;Y) = -\sum P_X \log P_X(x) - H(X/Y)$$

$$= \sum_{x \in X} \sum_{y \in Y} P_{XY}(x,y) \log \frac{P_{XY}(x,y)}{P_X(x) \cdot P_Y(y)}$$

Definition 6.2. The **joint entropy** $H(X,Y)$ of two random variables X and Y is given as:

$$H(X,Y) = -\sum_{\text{all } x\in X}\sum_{\text{all } y\in Y} p_{XY}(x,y)\log_2 p_{XY}(x,y)$$

§6.2 Conditional entropy

You've seen conditional probability. Obviously, we'll make you see conditional entropy.

Definition 6.3. Consider two random variables X and Y with joint PMF $p_{XY}(x,y)$. Then the **conditional entropy** of Y given X is represented as:

$$H(Y|X) = \sum_{\text{all } x\in X} p_X(x)H(Y|X=x)$$

$H(Y|X=x) = -\sum_{y\in Y} P_{Y|X}(y|x)\log P_{Y|X}(y|x)$

$\Rightarrow H(Y|X) = -\sum_{x\in X} P_X(x)\sum_{y\in Y} P_{Y|X}(y|x)\log P_{Y|X}(y|x) = -\sum_{x\in X}\sum_{y\in Y} P_X(x)P_{Y|X}(y|x)\log P_{Y|X}(y|x)$ ($P_X(x)P_{Y|X}(y|x) = P_{XY}(x,y)$)

Problems

1. Can you simplify the expression into a nice double summation? If so, can you then write that as the expectation of some RV (the second part is not compulsory).
2. Prove that
$$H(X,Y) = H(X) + H(Y|X)$$

$H(X,Y) = -\sum_{x\in X}\sum_{y\in Y} P_{XY}(x,y)\log P_{Y|X}(y|x)$

$H(X,Y) = -\sum\sum P_{XY}(x,y)\log P_{XY}(x,y)$

$= -\sum\sum P_{XY}(x,y)\log(P_X(x)\times P_{Y|X}(y|x)) = H(X)+H(Y|X)$

§6.3 Mutual Information

Let's make a langrangian style twist on the previous problem.

Definition 6.4. Consider two random variables X and Y with joint PMF $p_{XY}(x,y)$ and marginal or standard PMFs $p_X(x)$ and $p_Y(y)$. Then we define the mutual information of X and Y as:

$$I(X;Y) = H(X) - H(X|Y)$$

It seems that the mutual entropy depends on the order of X and Y. But does it really? We shall soon see.

Problems

1. Represent the mutual information expression in terms of summations involving the PDFs only.
2. Looking at the expression, can you comment something about the order of X and Y. Can you give an expression for $I(X;Y)$ in terms of entropies which embodies this above fact directly (2nd question is optional)?

   ↳ doesn't matter

   $I(X;Y) = H(X) - (H(X,Y) - H(Y))$

   $= H(X) + H(Y) - H(X,Y)$
3. What can we say about $I(X;X)$?

   $H(X)$